

Visual Deep Reinforcement Learning for Adaptive Traffic Signal Control: A Neuro-Symbolic XAI Approach

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Abstract—Deep Reinforcement Learning (DRL) offers promising solutions for Adaptive Traffic Signal Control. However, visual DRL models that map raw pixels directly to control actions operate as “black boxes,” hindering their deployment in safety-critical infrastructure due to a lack of interpretability. This paper presents a visual Deep Q-Network agent trained in a high-fidelity CARLA-SUMO co-simulation environment, capable of reducing vehicle waiting times by approximately 23% compared to fixed-time controllers. Beyond performance, we introduce a novel Neuro-Symbolic Explainable AI framework to validate the agent’s decision-making. We employ SHAP for visual introspection and train symbolic surrogate models (Decision Trees and Explainable Boosting Machines) to extract human-readable logic rules from the neural policy. Results demonstrate that the agent implicitly learns to estimate traffic density from photorealistic images, with symbolic models achieving high fidelity in approximating the visual policy. This approach effectively bridges the gap between the performance of deep learning and the transparency required for real-world traffic management.

I. INTRODUCTION

Urban traffic congestion remains one of the most critical challenges for modern cities, resulting in severe economic losses, increased carbon emissions, and reduced quality of life [1], [2]. According to United Nations projections [3], 68% of the world’s population will live in urban areas by 2050, exacerbating mobility issues if infrastructure management does not evolve. Traditional traffic signal control (TSC) systems [4], predominantly based on fixed-time schedules or simple actuation rules, often fail to adapt to dynamic and non-linear traffic flows, leading to inefficient intersection management.

In recent years, Deep Reinforcement Learning (DRL) has emerged as a promising paradigm for Adaptive Traffic Signal Control (ATSC). DRL agents can learn complex control policies directly from interaction with the environment, aiming to maximize long-term rewards such as traffic throughput or minimized waiting times. However, the majority of existing research relies on state representations derived from inductive loops or abstract matrices (e.g., position/velocity vectors), which requires expensive sensor infrastructure and simplifies the visual complexity of real-world scenarios.

Emerging approaches propose utilizing visual data (camera feeds) as the primary input for DRL agents, leveraging Convolutional Neural Networks (CNNs) to extract

features directly from raw pixels. While this approach offers high scalability and lower installation costs, it introduces a significant challenge: the “Black Box” problem. End-to-end learning models, mapping pixels directly to traffic light phases, lack transparency. In safety-critical domains like traffic management, authorities and operators require not only performance but also interpretability. Deploying a neural network that controls critical infrastructure without understanding the rationale behind its decisions is often unacceptable.

To address this gap, this work presents a comprehensive framework for visual ATSC that combines high-fidelity simulation with a novel Neuro-Symbolic Explainable AI (XAI) approach. We develop an autonomous agent based on the Deep Q-Network (DQN) algorithm, trained within a co-simulation environment integrating CARLA (for photorealistic visual perception) and SUMO (for microscopic traffic logic). Unlike previous works, we do not stop at performance evaluation. We propose a post-hoc explainability methodology that employs symbolic surrogate models (Decision Trees and Explainable Boosting Machines) to extract logical rules from the agent’s behavior, alongside visual introspection techniques (SHAP) to validate feature relevance.

The main contributions of this paper are summarized as follows:

- Development of a high-fidelity co-simulation environment (CARLA-SUMO) that enables training DRL agents using exclusively photorealistic images, bridging the gap between simulation and reality.
- Implementation and validation of a visual DQN agent capable of outperforming traditional fixed-time controllers in diverse traffic density scenarios.
- Introduction of a Neuro-Symbolic XAI framework that validates the agent’s decision-making process. This includes visual attribution methods to ensure the model focuses on relevant traffic features (e.g., queues vs. background noise) and the extraction of human-readable logic rules via surrogate modeling to interpret the learned policy.

The remainder of this paper is organized as follows: Section II reviews related work in visual RL and XAI for traffic. Section III details the system architecture and the DRL methodology. Section IV describes the proposed Neuro-Symbolic XAI framework. Section V presents the experimental results and the interpretability analysis, and Section VI concludes the paper.

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II. RELATED WORK

A. DRL for Traffic Signal Control

Traditional ATSC systems, such as SCATS [5] and SCOOT [6], rely on manually engineered models and extensive sensor loop infrastructure. In contrast, DRL [7] has demonstrated superior capability in handling non-linear traffic dynamics without requiring explicit physical modeling. Early approaches utilized tabular Q-learning [8], but the high dimensionality of traffic states led to the adoption of Deep Q-Networks [9]. Works such as [10], [11] and specialized variations like 3DQN [12] have successfully reduced waiting times compared to fixed-time controllers. However, the majority of these studies define the state space S using abstract feature vectors (e.g., queue lengths, vehicle position matrices) extracted directly from the simulator's internal memory, assuming perfect observability and sensing capabilities that are costly to replicate in real-world scenarios.

B. Visual Perception in RL

To overcome the dependency on inductive loops and simplify state representation, recent research has explored visual perception as the primary input for control agents. This “end-to-end” approach utilizes Convolutional Neural Networks (CNNs) to map raw pixel data to control actions. While promising, many existing visual approaches, such as [13], rely on low-fidelity simulators or grid-based representations that lack the visual complexity of urban environments. Our work addresses this by utilizing a high-fidelity co-simulation between CARLA [14] and SUMO [15] where the agent must learn robust features from photorealistic images, significantly narrowing the simulation-to-reality gap.

C. Explainable AI in Traffic Control

As DRL models become more complex, their “black-box” nature poses a barrier to adoption in safety-critical infrastructure. Explainable AI techniques [16], [17] are increasingly sought to audit autonomous agents. In the domain of computer vision, pixel-attribution methods like saliency maps or SHAP (Shapley Additive Explanations) [18] are common. However, in the specific niche of ATSC, XAI is largely underexplored, particularly for visual agents. Most interpretability efforts focus on vector-based agents, analyzing weight matrices or attention mechanisms [19]. There is a lack of neuro-symbolic approaches that bridge the gap between sub-symbolic perception (pixels) and symbolic logic (traffic rules). This paper contributes to this area by not only inspecting visual attention but also extracting symbolic surrogate models to validate the logic behind the visual agent's decisions.

III. SYSTEM ARCHITECTURE & METHODOLOGY

The proposed system integrates a high-fidelity visual simulation environment with a DRL agent designed to minimize traffic congestion based purely on visual inputs.

A. Co-Simulation Environment

To bridge the gap between visual realism and traffic modeling accuracy, we developed a co-simulation framework linking CARLA and SUMO simulators.

- **Visual Layer (CARLA):** Renders the 3D scene, providing photorealistic textures, lighting, and occlusions. It acts as the “camera sensor,” generating the raw pixel input for the agent.
- **Logic Layer (SUMO):** Manages microscopic traffic dynamics, vehicle routing, and physical interactions. It ensures valid traffic flows and provides ground-truth metrics used for reward calculation and XAI validation.

This hybrid approach allows the agent to train in a visually complex environment while maintaining the stochastic traffic patterns characteristic of real urban intersections.

B. MDP Formulation

The TSC problem is formulated as a Markov Decision Process (MDP) [20] defined by the tuple (S, A, R, γ) :

- 1) **State Space (S):** The state s_t is defined exclusively by an image of the intersection captured from an RGB camera. Based on our hyperparameter optimization, the raw camera feed is resized to 100×100 pixels and converted to a single channel. This configuration was found to maximize computational efficiency without losing critical spatial information.
- 2) **Action Space (A):** The action space is discrete. The agent selects a phase a_t from the set of available green phases for the intersection which keeps active for 7 seconds. If the selected action differs from the current phase, a yellow transition phase of 3 seconds is enforced by the system to ensure safety constraints.
- 3) **Reward Function (R):** The objective is to minimize the total cumulative waiting time of all vehicles. The base reward r_t is calculated as the negative sum of the waiting time $w_{i,t}$ of every vehicle i currently in the intersection, as shown in Equation 1. To stabilize training and handle extreme values during congestion, we apply a logarithmic transformation.

$$r_t = -\log\left(1 + \sum_{i=1}^{N_t} w_{i,t}\right) \quad (1)$$

This shaped reward r_t encourages the agent to clear queues quickly while preventing gradient explosions.

C. Deep Q-Network Architecture

We implement a Deep Q-Network [9] to approximate the optimal action-value function $Q^*(s, a)$. The network architecture is a CNN designed to process spatial dependencies in the traffic images. It consists of three convolutional layers with ReLU activation, followed by a flattening layer and two fully connected layers. This design omits pooling layers to preserve the precise spatial location of vehicles, which is critical for determining specific lane occupancy. The network is trained using *experience replay* to break temporal correlations and a *target network* to stabilize learning updates.

Figure 1 shows the complete architecture of the system and environment developed.

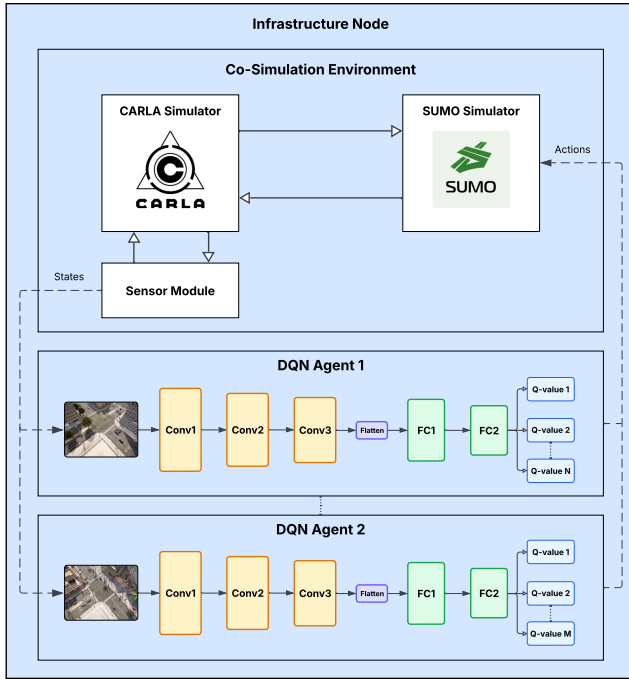


Fig. 1. System architecture.

IV. NEURO-SYMBOLIC EXPLAINABLE AI FRAMEWORK

While the DQN agent demonstrates effective control, its reliance on deep CNNs makes it opaque. To validate that the agent learns meaningful traffic concepts rather than exploiting visual artifacts, we introduce a dual-layer XAI framework: Visual Introspection (Pixel-level) and Neuro-Symbolic Surrogate Modeling (Logic-level).

A. Visual Introspection

To understand which regions of the input image influence the agent's Q-values, we employ SHAP [18]. Specifically, we use a gradient-based explainer suitable for deep networks. For a given input state s , the method assigns an attribution value to each pixel, representing its contribution to the selected action a . This visual validation allows us to verify if the agent focuses on relevant areas (e.g., lanes, vehicles) and ignores background noise (e.g., buildings, shadows).

B. Neuro-Symbolic Surrogate Modeling

Visual heatmaps show where the agent looks, but not what logic it applies. To extract explicit decision rules, we treat the trained DQN agent as a "black-box oracle" and train interpretable white-box models to mimic its behavior using symbolic features.

1) *Symbolic Dataset Generation*: We run the trained DQN agent for 25 episodes under varying traffic conditions (low density and high density). During these runs, we extracted a synchronized dataset $\mathcal{D} = \{(x_{sym}^{(i)}, y_{dqn}^{(i)})\}$, where:

- $y_{dqn}^{(i)}$: The action selected by the DQN agent at step i .
- $x_{sym}^{(i)}$: A vector of symbolic attributes extracted from the simulation ground truth (SUMO), including the current active phase, waiting times and cumulative vehicle counts for each incoming lane.

2) *Surrogate Models*: We trained two types of interpretable models on dataset $\mathcal{D}$:

- **Decision Tree (DT)**: A shallow tree to extract explicit IF-THEN rules representing the agent's policy.
- **Explainable Boosting Machine (EBM) [21]**: A model that combines gradient boosting with Generalized Additive Models (GAMs) [22] to provide global feature importance scores, quantifying which symbolic attributes have the highest impact on the agent's decisions.

3) *Fidelity Evaluation*: The validity of these explanations depends on the fidelity score, defined as the accuracy of the surrogate model in predicting the DQN's actions on a hold-out test set. A high fidelity score implies that the logical rules extracted from the surrogate accurately represent the underlying policy of the neural network.

V. EXPERIMENTS AND RESULTS

A. Control Performance Evaluation

Before analyzing interpretability, we validated the control performance of the proposed Visual DQN agent. We compared it against a standard fixed-time controller in two different intersections under two traffic conditions: *real traffic* (low density) and *saturated traffic* (high density). As shown in Figure 2, one intersection, hereafter referred to as *complex*, consists of four incoming lanes controlled by four traffic lights, while the other, referred to as *simple*, consists of three incoming lanes and three corresponding traffic lights.

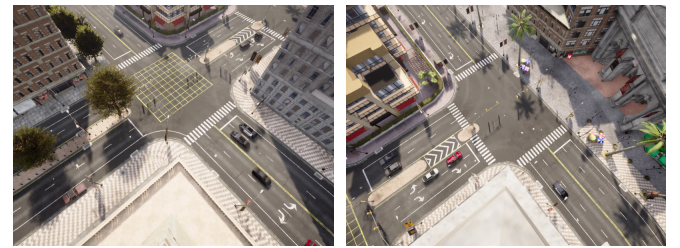


Fig. 2. Simulated intersections in CARLA.

Figure 3 summarizes the results obtained. The DQN agent achieved a significant reduction in average waiting time (Avg WT) across all scenarios.

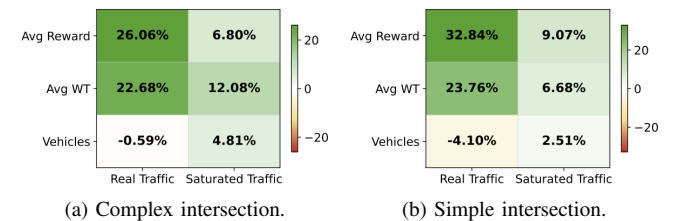


Fig. 3. Improvements of the DQN agent over the fixed-time TSC system.

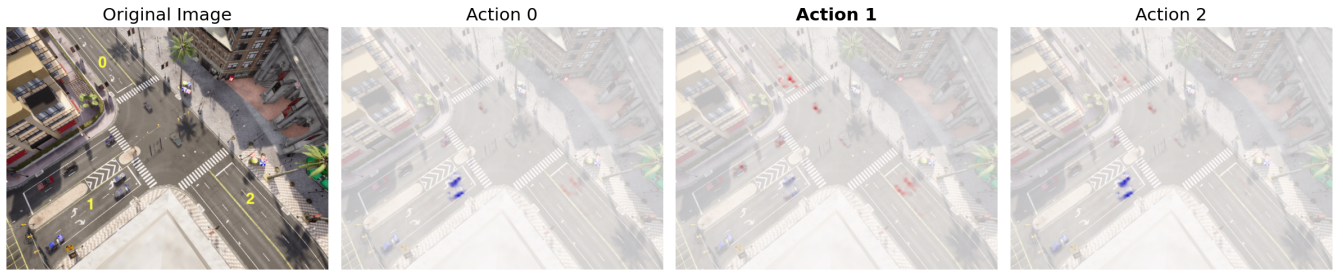


Fig. 4. Example of visual explainability using SHAP values on an input image of the *simple* intersection. In the original image, the correspondence between each traffic light and its associated action is indicated in yellow, and the action selected by the agent is indicated in bold.

In the realistic scenario, the agents of both intersections reduced waiting times by approximately 23% compared to the baseline. Even in saturated conditions, where optimization margins are tighter, the agent maintained superior performance. These results confirm that the pixel-based policy effectively manages complex traffic flows.

B. Visual Explainability

To analyze the agent's decision-making process, SHAP's Gradient Explainer was applied to the images provided as input. Figure 4 presents an illustrative example, showing first the original (unprocessed) input image, followed by the corresponding SHAP value maps for each possible action. In these maps, red regions indicate positive contributions that increase the likelihood of selecting a given action, whereas blue regions indicate negative contributions that decrease it.

The results show a consistent pattern across actions. For each action, the agent assigns positive importance to the absence of vehicles in the lanes associated with the other actions, while the presence of vehicles in those lanes contributes negatively. In other words, instead of explicitly rewarding the presence of vehicles in the lane corresponding to the selected action, the agent primarily relies on penalizing alternative actions when their lanes are occupied. This suggests that the agent's policy is driven more by the relative disadvantage of competing actions than by direct evidence favoring the chosen one.

Additionally, for actions 1 and 2, the SHAP maps reveal that vehicle motion within a lane, indicating that its traffic light is active, has a positive influence on selecting a different action. This indicates that the agent has learned to interpret moving vehicles as a signal that the corresponding lane is already being served, and therefore that another action should be selected instead.

C. Neuro-Symbolic Fidelity Analysis

To bridge the visual policy with symbolic logic, we trained the surrogate models (DT and EBM) on the dataset $\mathcal{D}$ generated from the agent's actions. The first step is to evaluate the fidelity score, i.e., how accurately these white-box models mimic the DQN's decisions.

Both the DT and the EBM achieved a fidelity accuracy of 85% on the *simple* intersection but a fidelity of 72% on the *complex* intersection. These results indicate that the

complex non-linear mapping from pixels to actions learned by the CNN can be effectively approximated by symbolic rules based on queue lengths. This implies the agent has implicitly learned to "count" vehicles visually. However, at complex intersections, the symbolic model fails to faithfully represent the agent's decision policy, leading to a noticeable drop in fidelity.

D. Global Interpretability

Using the Explainable Boosting Machine, we analyzed the global feature importance. As hypothesized, the symbolic features with the highest impact on the decision were the number of vehicles of each lane. Additionally, the EBM allows extracting the contribution of each attribute to the selection of each class according to its different values. Figure 5 shows an example for the number of vehicles in Lane 2 of the *simple* intersection.

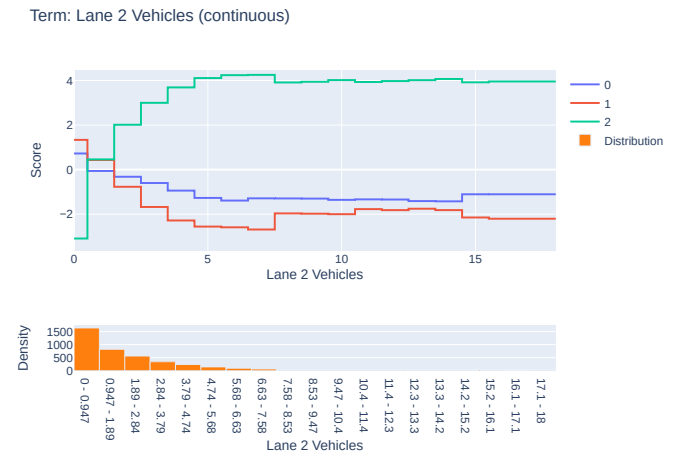


Fig. 5. Global contribution of the number of vehicles in Lane 2 to the prediction of actions 0, 1, and 2.

It can be observed that, starting from two vehicles in Lane 2, action 2 becomes predominant, i.e., setting the traffic light of Lane 2 to green. Moreover, as the number of vehicles in Lane 2 increases, the probability of choosing action 2 rises, while the probabilities of selecting actions 0 and 1 decrease.

Furthermore, the Decision Tree allowed us to extract explicit logic rules, as shown in Figure 6.

These rules align perfectly with human traffic management logic: if two lanes have low demand or they are empty and

Extracted Decision Rules (Fidelity: 85.30%)

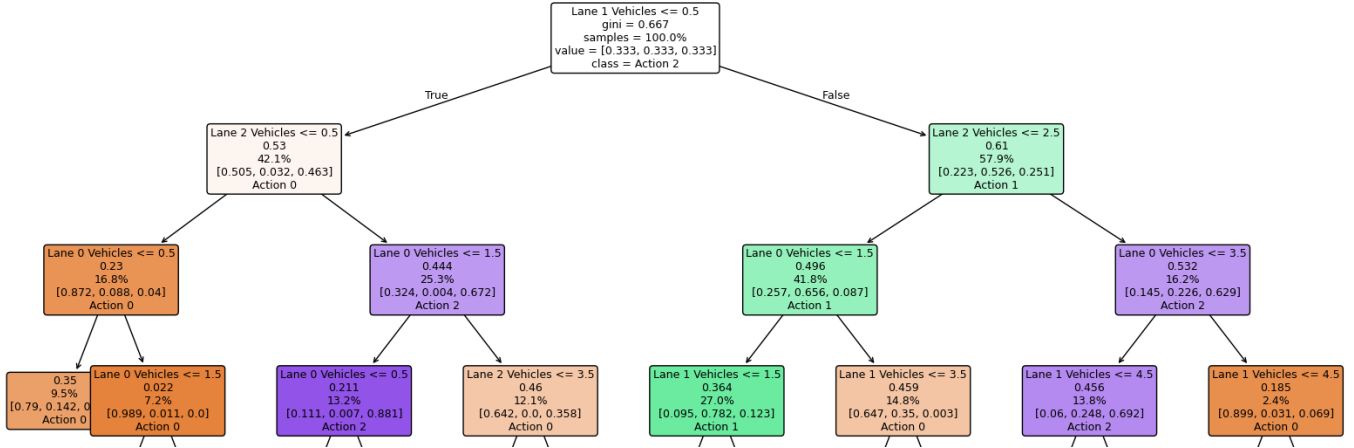


Fig. 6. Decision tree rules extracted from the *simple* intersection agent policy.

other lane is congested, the agent decides to turn green the traffic light of the congested one. This behavior of the agent is consistent with that analyzed in the SHAP visualizations in Section V-B, where it is observed that the agent assigns greater importance to empty lanes and uses the vehicles at a given traffic light to reduce the probability of the remaining actions.

The derivation of such semantically meaningful rules from a purely visual agent validates the safety and coherence of the learned policy.

E. Local Explainability

Finally, we conducted a local analysis on specific test instances to assess whether the EBM's local explanations are meaningful and coherent.

Figure 7 illustrates a case in which the agent's decision agrees with that of the symbolic model. In this example, the agent receives the input image and selects action 0. Similarly, the symbolic model also predicts action 0 based on the number of vehicles per lane extracted from the input scene. The plot on the right shows the contribution of each feature to the prediction of each action.

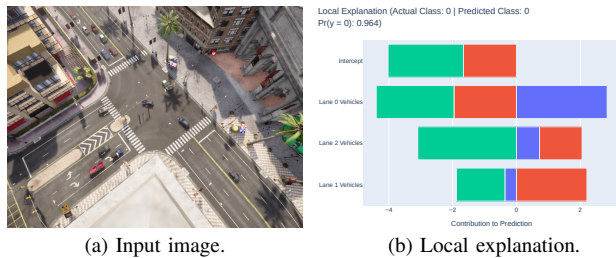


Fig. 7. Explanation of correct symbolic model prediction.

Since Lane 0 has five vehicles waiting, Lane 1 has three, and Lane 2 has none because it is currently open, both the agent and the symbolic model decide to switch the traffic light to Lane 0. The symbolic explanation shows that the number of vehicles in Lanes 0 and 1 contributes positively

to their corresponding action predictions, while contributing negatively to the prediction of the action associated with Lane 2. In particular, the high number of vehicles in Lane 0 reduces the relative contribution of Lane 1, ultimately making action 0 the selected decision.

In contrast, Figure 8 shows a case in which the symbolic model's decision differs from that of the agent. The agent selects action 2, whereas the EBM predicts action 0. However, as shown by the local explanation, the predicted probabilities are very close (0.52 for action 0 and 0.48 for action 2), indicating uncertainty in the model's decision. This uncertainty arises because Lane 0 has four waiting vehicles, while Lane 2 has five. These results suggest that most of the symbolic model's errors stem from the use of overly simple features, which fail to capture all the relevant details influencing the agent's decision.



Fig. 8. Explanation of incorrect symbolic model prediction.

VI. CONCLUSIONS AND FUTURE WORK

This work presented a comprehensive framework for visual Adaptive Traffic Signal Control, bridging the gap between high-performance Deep Reinforcement Learning and the interpretability requirements of safety-critical infrastructure. By leveraging a high-fidelity co-simulation environment using CARLA and SUMO, we successfully trained a DQN agent capable of making optimal control decisions based solely on raw photorealistic images, achieving a reduction in waiting times of approximately 23% compared to traditional fixed-time controllers in diverse traffic scenarios.

Beyond performance, the primary contribution of this study is the novel Neuro-Symbolic XAI framework designed to audit the “black-box” nature of visual agents. The application of SHAP revealed that the agent’s convolutional layers correctly prioritize semantic regions of interest, specifically, vehicle queues and active lanes, while ignoring visual noise such as buildings or vegetation. Furthermore, the use of symbolic surrogate models allowed us to extract human-readable logic rules from the agent’s behavior. The high fidelity scores obtained (up to 85%) demonstrate that the agent implicitly learns to “count” vehicles and estimate densities from pixels, aligning its latent policy with logical traffic management principles.

However, the fidelity drop observed in complex intersections (72%) highlights that simple symbolic features, such as vehicle counts, may not fully capture the nuanced spatial-temporal reasoning of the CNN in high-complexity scenarios.

Future work will focus on three main directions:

- **Enhanced symbolic representation:** Improving the surrogate models by incorporating richer symbolic features to better approximate the agent’s policy in complex intersections.
- **Adversarial robustness:** Analyzing the visual agent’s resilience against adversarial attacks or severe weather conditions (e.g., heavy rain, occlusion) using the proposed XAI framework to detect failure modes.
- **Human-in-the-loop validation:** Integrating these explanations into a real-time dashboard to evaluate whether the extracted rules and visual heatmaps effectively assist human operators in trust-calibration and decision-making tasks.

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